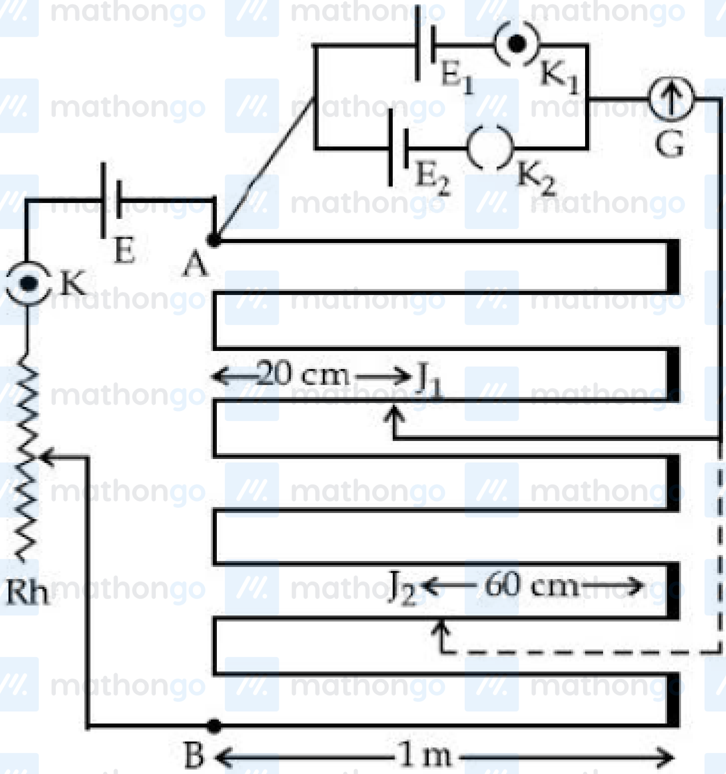
Q1: 25 Feb (Shift 1) - Numerical

In the given circuit of potentiometer, the potentital difference E across $AB(10 \mathrm{~m}$ length $)$ is larger than E_1 and E_2 as well. For key K_1 (closed), the jockey is adjusted to touch the wire at point J_1 so that there is no deflection in the galvanometer. Now the first battery $\left(E_1\right)$ is replaced by second battery $\left(E_2\right)$ for working by making K_1 open and E_2 closed. The galvanometer gives then null deflection at J_2. The value of $\frac{\mathrm{E}_1}{\mathrm{E}_2}$ is $\frac{\mathrm{a}}{\mathrm{b}}$, where $\mathrm{a}=$



Q2: 26 Feb (Shift 1) - Single Correct

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : An electron microscope can achieve better resolving power than an optical microscope.

Reason R : The de Broglie's wavelength of the electrons emitted from an electron gun is much less than wavelength of visible light. In the light of the above statements, choose the correct answer from the options given below:

(1) A is true but R is false.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) Both A and R are true and R is the correct explanation of A.

(4) A is false but R is true.

Answer Key

Q1 (1)
Q2 (3)

Q1 (1)

$\frac{E_1}{E_2} = \frac{I_1}{l_2}$

$= \frac{3 \times 100 \text{ cm} + (100-20)\text{cm}}{7 \times 100 \text{ cm} + 60 \text{ cm}}$

$= \frac{380}{760} = \frac{1}{2} = \frac{\mathrm{a}}{\mathrm{b}}$

$\mathrm{a} = 1$

Q2 (3)

Resolution limit $(\Delta\theta) = \frac{1.22\lambda}{\mathrm{d}}$

Resolution power $= \frac{1}{\text{Resolution limit}}$

$\lambda \downarrow \Delta\theta \downarrow$

$\Delta\theta \downarrow$ Power $\uparrow$